



Assiut University
Faculty of Computers & Information
Computer Science Department



Intelligent Systems

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Grading

Assessment	Grade %
Final Exam	50%
Class Work	15%
Mid-Term Exam	15%
Project	20%

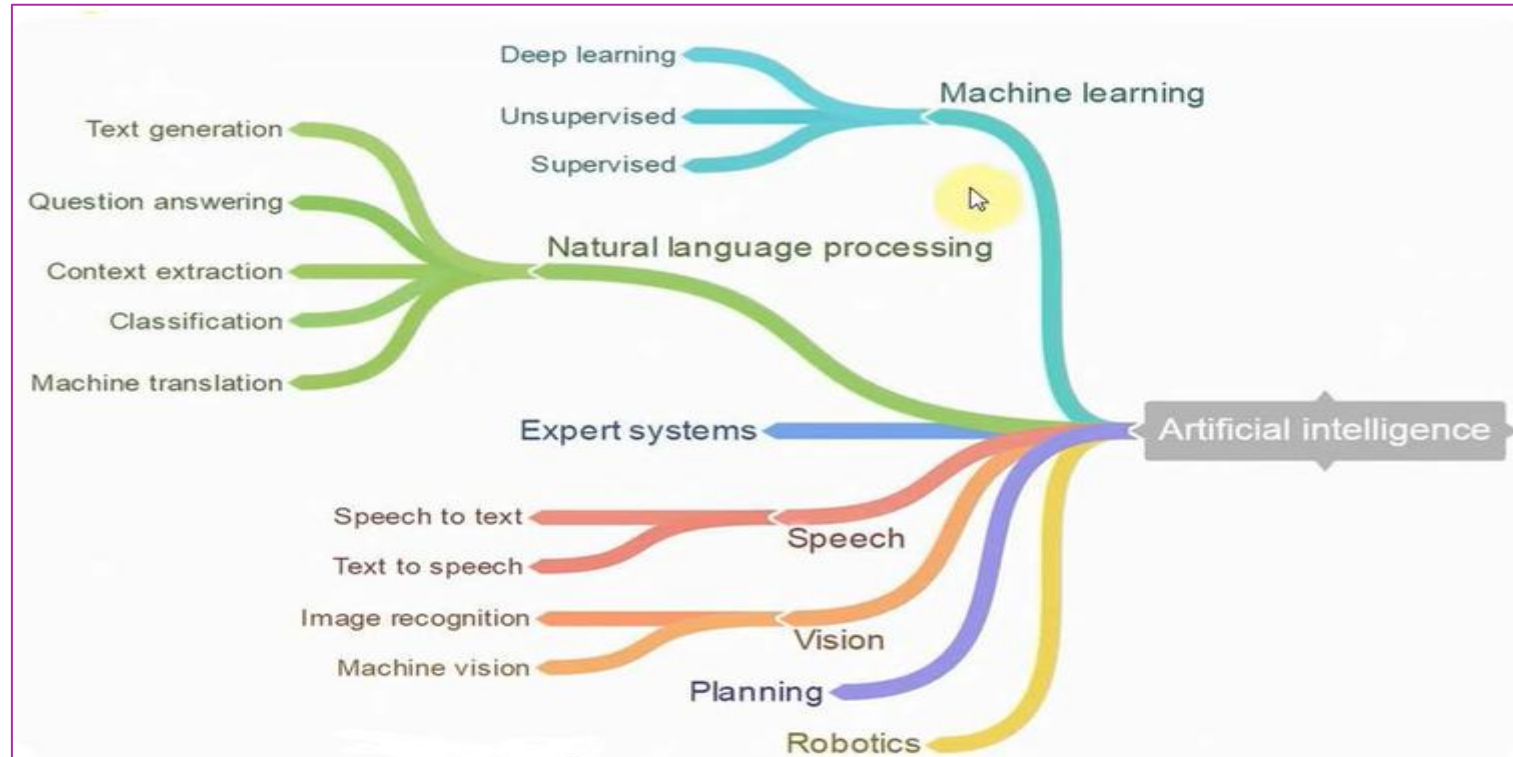
We call ourselves **Homo sapiens**—man the wise—because our intelligence is so important to us.

The field of Artificial Intelligence, or AI: is concerned with not just understanding but also building intelligent entities—machines that can compute how to act effectively and safely in a wide variety of novel situations.

The ultimate achievement in AI would be to construct a machine that can **mimic** or exceed human mental capabilities, including :

- ❑ Reasoning
- ❑ Under standing
- ❑ Imagination
- ❑ Recognition
- ❑ Creativity, and Emotions.

Where we can see AI ?



ARTIFICIAL INTELLIGENCE

techniques that enable computers to mimic human intelligence

MACHINE LEARNING

*techniques where machines improve
at tasks with experience*

DEEP LEARNING

*machine learning using deep neural networks
(inspired by human brain architecture)*

Applications of AI



- AI has been applied in many fields:
 - Entertainment (e.g., game playing).
 - Internet (e.g., semantic search engines, spam fighting).
 - Robotics (e.g., driverless vehicles, vacuum cleaners).
 - Military (e.g., logistics planning).
 - Space (e.g., rovers, autonomous planning/scheduling).
 - Medical (e.g., expert systems for medical diagnosis).
 - Flight (e.g., auto-pilot).
 - Finance (e.g., stock market prediction, fraud detection).
 - Weather forecasting.

What is AI ?

- Some definitions from literature:
 1. The exciting of new effort to make computers **think** ... machines with **minds**, in the full and literal sense.
 2. The art of creating machines that **perform** functions that require intelligence when performed by **people**.
 3. The study of **mental faculties** through the use of **computational models**.
 4. AI ... is concerned with **intelligent behavior** in artifacts.

What is AI ?

Different people approach **AI** with different goals in mind. Two important questions to ask are: Are you concerned with **thinking** or **behavior**? Do you want to **model humans** or **work from an ideal standard**?

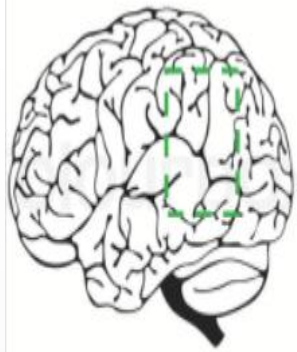
	Human	Rationally
Think	Systems that think like humans .	Systems that think rationally .
Act	Systems that act like humans .	Systems that act like rationally .

	Success in terms of human performance	Success in terms of rational performance
Mimic thought process and reasoning	Think Humanly	Think Rationally
Mimic Behavior	Act Humanly	Act Rationally
	More to do with hypothesis & experiments	More to do with mathematics & engineering

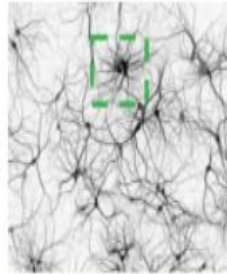


Think Humanly

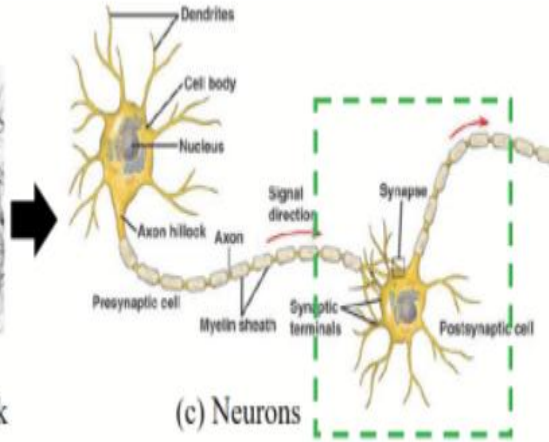
- Cognitive modeling approach
 - If we want the computer to think as humans, we must first determine how humans think.
 - An interdisciplinary field of **cognitive science**.
 - It brings together models from AI and experimental techniques from psychology.
 - It tries to come up with precise and testable theories of how the human mind works.
 - Example: Neural Networks.



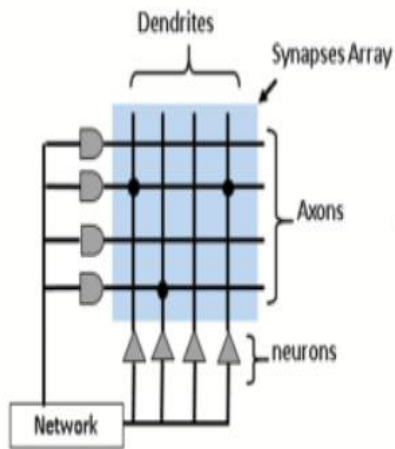
(a) Brain



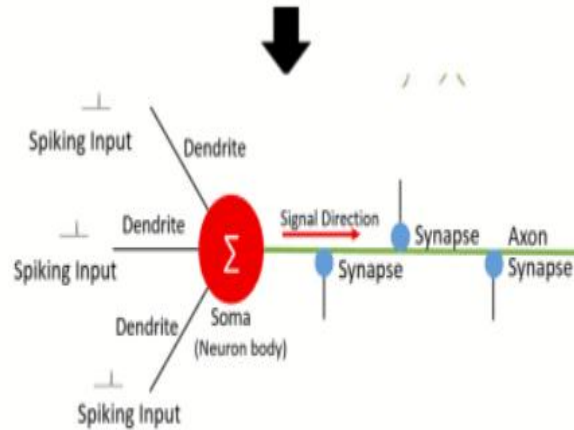
(b) Neuron network



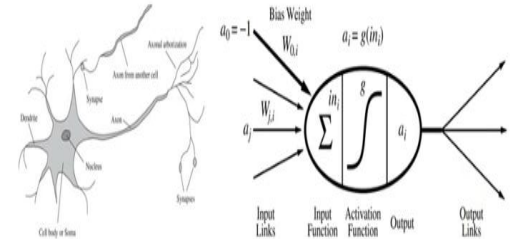
(c) Neurons



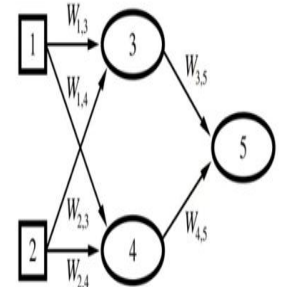
(e) Neuron Network



(d) Neuron Structure



Neural Network
is a set of neuron
units.



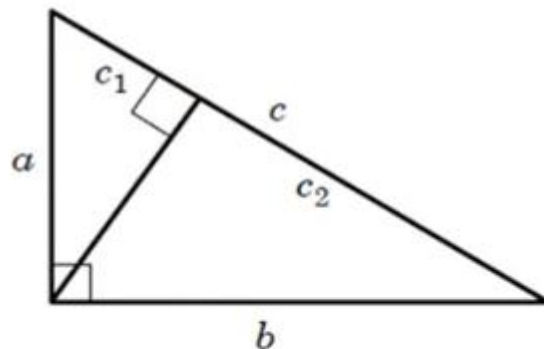
Think Rationally

- “Laws Of Thought” approach:
 - Uses logic to reach a decision given some facts via logical inferences.
 - Greek philosopher Aristotle was the first to try to codify the way of thinking.
 - His deductive reasoning always gave correct conclusions when given correct premises.
 - Example: “All Cats are Pets; Bob is a Cat; therefore, Bob is a Pet”,
 - Example: Theorem proving in first-order logic.

THINK RATIONALLY: "LAWS OF THOUGH" APPROACH

○ Theorem proving:

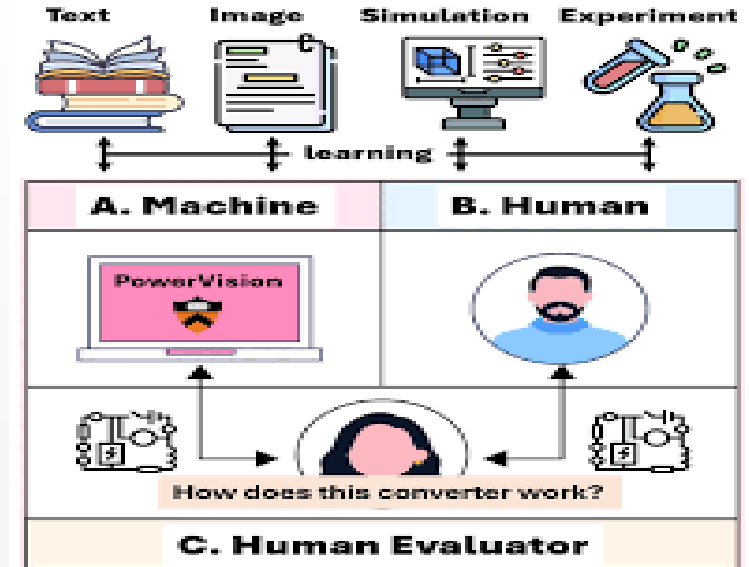
- Set of rules:
 - R1: $X = Y / Z \Leftrightarrow X.Z = Y$
 - R2: $X = Y \Leftrightarrow X + Z = Y + Z$
 - R3: $X * Y + X * Z \Leftrightarrow X * (Y + Z)$
- Set of facts (premises):
 - F1: $b / c \Leftrightarrow c_2 / b$
 - F2: $a / c \Leftrightarrow c_1 / a$
 - F3: $c = c_1 + c_2$
- Goal:
 - Show that: $c^2 = a^2 + b^2$



Act
Humanly

Turing test

The Turing test, proposed by Alan Turing (1950), was designed as a thought experiment that would sidestep the philosophical vagueness of the question “Can a machine think?” A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.



- Turing Test Approach

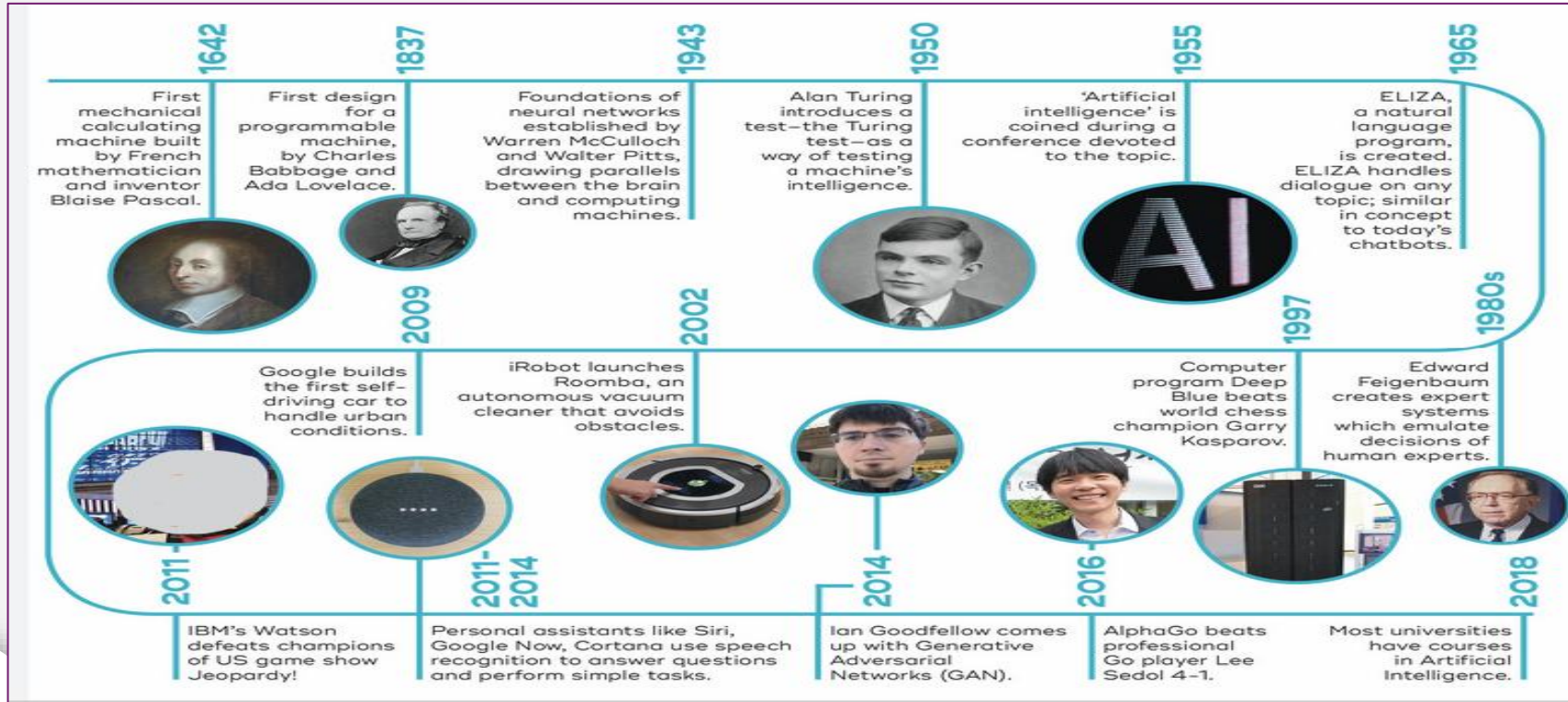
- To pass Turing test, computer needs:
 - Natural language processing
 - Knowledge representation
 - Automated reasoning
 - Machine learning
 - Total Turing Test: interrogator can pass physical objects to the computer (through a hatch).
 - To pass Total Turing test, computer needs:
 - Computer vision
 - Robotics
- 



Act Rationally

- Rational Agent Approach
 - Perceive the environment, and act so as to achieve one's goal.
 - Not necessary to do the best action:
 - There is not always an absolutely best action.
 - There is not always time to find the best action.
 - An action that's good enough can be acceptable.
 - Example: Game playing.
 - Sample approach: Tree-searching strategies.
 - Problem: Choosing what to do given the constraints.
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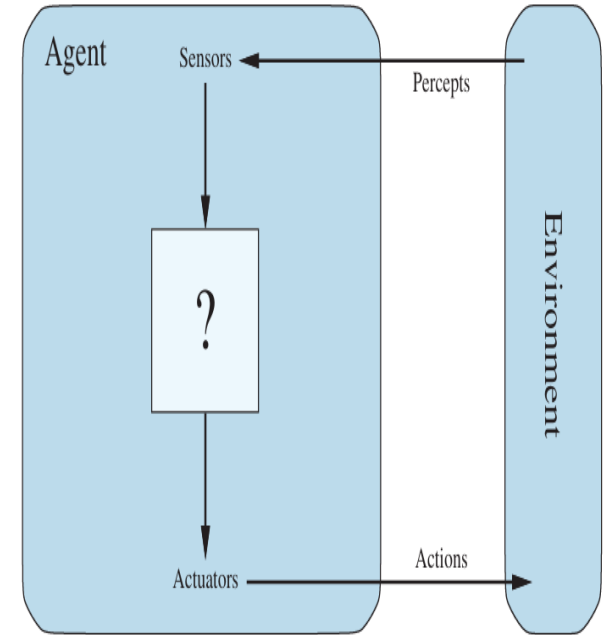
History of Artificial Intelligence



What is an AI Agent?

- **An agent** is just something that acts (agent comes from the Latin agere, to do).
- **A rational agent** is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.
- **Intelligence** is concerned mainly with rational action. Ideally, an intelligent agent takes the best possible action in a situation.

- **An agent** is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.
- Examples:
 - ✓ **A human agent** has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators.
 - ✓ **A robotic agent** might have cameras and infrared range finders for sensors and various motors for actuators
 - ✓ **A software agent** receives file contents, network packets and human input (keyboard/mouse/touchscreen/voice) as sensory inputs and acts on the environment by writing files, sending network packets, and displaying information or generating sounds.



Agents interact with environments through sensors and actuators.



Human Agent

Sensors Eyes, ears, nose, skin,...

Actuators Hands, legs, mouth,...



Robotic Agent

Sensors Cameras, infrared ,...

Actuators Various motors, wheels,...



A software Agent

Sensors Keystrokes, file contents, received network ...packages

Actuators ,displaying on the screen, writing files ... ,sending network packets

- **percept** to refer to the content an agent's sensors are perceiving.
- **Percept sequence** agent's percept sequence is the complete history of everything the agent has ever perceived
- **Agent function** maps any given percept sequence to an action.

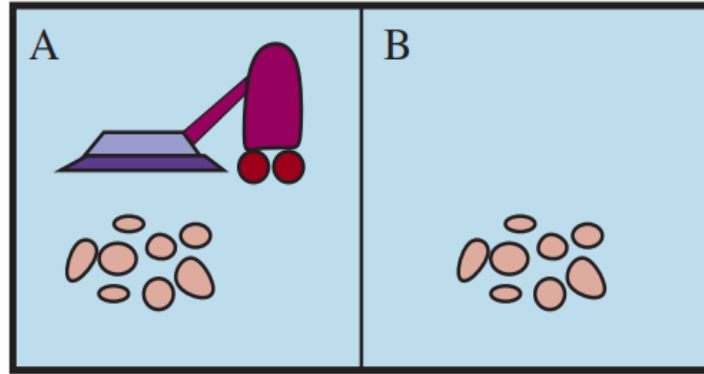
➤ An agent's behavior is described by the agent function which maps from percept histories to actions:

$$f: P^* \rightarrow A$$

➤ **Agent function** will be implemented by an **agent program** which runs on the physical architecture to produce **f**

$$\text{agent} = \text{architecture} + \text{program}$$

Robot Vacuum Cleaner



A vacuum-cleaner world with just two locations. Each location can be clean or dirty, and the agent can move left or right and can clean the square that it occupies. Different versions of the vacuum world allow for different rules about what the agent can perceive, whether its actions always succeed, and so on.

- ❖ **Percepts:** location and contents, e.g: [A,Dirty]
- ❖ **Actions:** Left, Right, Suck



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iRobot Roomba Model 4000 Red Robotic Vacuum Cleaner Tested Working. No battery.



rcw_7150 (97)

[100% positive](#) · [Seller's other items](#) · [Contact seller](#)



US \$99.96 or Best Offer

Condition: **Used** 

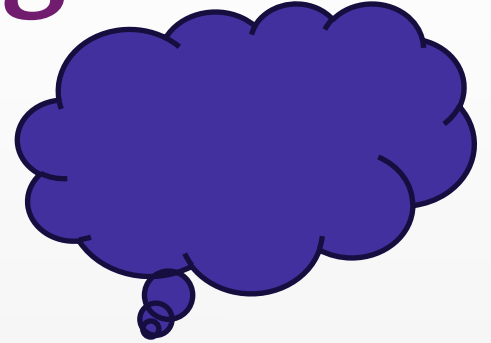
Buy It Now

[Add to cart](#)

[Make offer](#)

[♥ Add to Watchlist](#)

How to implement Agent function?



Agent function as look up Table:

❖ An agent actions is completely specified by the lookup table

❖ Drawbacks:

- Huge table
- Take a long time to build the table
- No **autonomy**

Percept sequence	Action
$[A, Clean]$	<i>Right</i>
$[A, Dirty]$	<i>Suck</i>
$[B, Clean]$	<i>Left</i>
$[B, Dirty]$	<i>Suck</i>
$[A, Clean], [A, Clean]$	<i>Right</i>
$[A, Clean], [A, Dirty]$	<i>Suck</i>
$\vdots$	$\vdots$
$[A, Clean], [A, Clean], [A, Clean]$	<i>Right</i>
$[A, Clean], [A, Clean], [A, Dirty]$	<i>Suck</i>
$\vdots$	$\vdots$

Rational Agent ?!

- A **rational agent** is one that does the right thing.
- In order to know the right thing, we need a performance measure.
- The **performance measure** is usually chosen by the agent designer.
- For each possible percept sequence, a rational agent should select an **action** that is expected to maximize the performance measure given the evidence provided by the percept sequence and whatever built-in **knowledge** the agent has.

For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Omniscience, learning, and autonomy

An **omniscient agent** knows the actual outcome of its actions and can act accordingly; but omniscience is **impossible in reality**

Key Characteristics

- **Perfect Knowledge:** An omniscient agent knows everything about the current state of the world.
- **Predictive Accuracy:** They know the precise outcome of any action they take.
- **Theoretical Nature:** Because they require perfect information and foresight, they are not practical for real-world AI, which often deals with uncertain or partially observable environments.
- **Distinction from Rationality:** A **rational agent** is not the same as an omniscient one. A rational agent maximizes *expected* performance based on its limited knowledge and perceptions, whereas an omniscient agent maximizes *actual* performance.

Specifying The Task Environment

- Task Environments are “problems” to which rational agents are the solutions!
- To specify a task environment, four items need to be defined:
 - Performance measure (GOAL)
 - Environment
 - Actuators
 - Sensors
- That is **PEAS** description
- Taxi Driver Example !

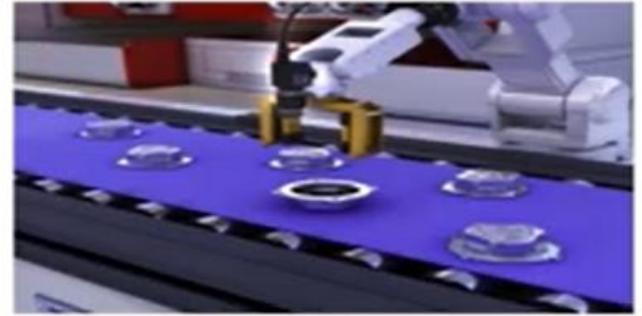
Agent Design (PEAS)

- ❖ **Performance:** How agent be assessed?
- ❖ **Environment:** What elements exists around agent?
- ❖ **Actuators:** How agent change the environment?
- ❖ **Sensors:** How agent sense the environment?

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Agent Design (PEAS)

Part-picking robot



- ❖ **Performance:** Percentage of parts in correct bins, speed
- ❖ **Environment:** Conveyor belt with parts, bins
- ❖ **Actuators:** Jointed arm and hand
- ❖ **Sensors:** Camera, joint angle sensors

Agent Design (PEAS)

Medical diagnosis system



- ❖ **Performance:** Healthy patient, minimize costs, lawsuits
- ❖ **Environment:** Patient, hospital, staff,.....
- ❖ **Actuators:** Screen display (questions, tests, diagnoses, treatments, referrals)
- ❖ **Sensors:** Keyboard (entry of symptoms, patient's answers)

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

Examples of agent types and their PEAS descriptions.

Assignment (1)

Design the agent for your projects?
(PEAS)

Environment Properties (ODESDA)

❖ Observable (or, partially observable)

An agent's sensors give it access to the complete state of the environment at each point in time

❖ Deterministic (or, stochastic)

The next state of the environment is completely determined by the current state and the action executed by the agent

❖ Episodic (or, sequential)

The agent's experience is divided into episodes, in each episode the agent receives a percept and then performs a single action

Observable
Deterministic
Episodic
Static
Discrete vs. continuous